

Yes. I went through the actual **Chapter 1 — “Introduction”** in your copy of John C. Hull, *Options, Futures, and Other Derivatives, 9th Global Edition*. Chapter 1 runs from printed pages **23–44** and contains sections 1.1–1.10. The goal below is that you can answer both conceptual questions and the calculation-style questions at the end of the chapter.

Chapter 1 — Complete Mastery Guide

The whole chapter can be understood around one central idea:

Derivatives are contracts whose value depends on something else, and people use them mainly to transfer risk, take risk, or exploit pricing inconsistencies.

That “something else” is called the **underlying asset or underlying variable**.

For example, an option on Google stock derives its value from Google’s stock price. A currency forward derives its value from an exchange rate. A weather derivative could depend on temperature or snowfall. Hull emphasizes that the underlying does not even have to be a traded financial asset.

1. What exactly is a derivative?

A **derivative** is a financial instrument whose value depends on another variable.

Imagine Google stock is:

$$S = \$100$$

I sell you a contract saying:

“Three months from now, you may buy Google from me for \$110.”

The value of that contract depends heavily on where Google’s stock price goes.

If Google rises to \$150, the right to buy at \$110 is valuable.

If Google falls to \$80, that right is almost useless.

So:

$\text{Value of derivative} = f(\text{underlying variable})$
--

Common underlyings include stocks, currencies, interest rates, commodities, stock indices, bonds, credit risk and even weather variables.

2. Why derivatives exist

This is one of the most important ideas of the entire chapter.

Three main groups use derivatives:

Trader	Goal
Hedger	Reduce existing risk
Speculator	Take risk hoping to profit
Arbitrageur	Exploit inconsistent prices to lock in profit

Almost everything in Chapter 1 comes back to these three.

Suppose EUR/USD might change.

A Moroccan or US company that must pay euros in three months may think:

“I don’t want to gamble on EUR/USD.”

That’s **hedging**.

A trader may think:

“I believe EUR will rise.”

That’s **speculation**.

Someone may discover EUR can simultaneously be bought for \$1.10 somewhere and sold for \$1.11 elsewhere.

That’s **arbitrage**.

3. Exchange-traded derivatives

Section 1.1 introduces **derivatives exchanges**.

An exchange provides standardized contracts.

For example, instead of you saying:

“I want exactly 82.37 ounces of gold delivered on November 14.”

the exchange defines things such as contract size, maturity dates, quality specifications and delivery rules.

This standardization makes contracts easier to trade.

The historical examples in Hull include the Chicago Board of Trade and Chicago Mercantile Exchange.

The clearing house

This concept is extremely important.

Imagine:

$$A \leftrightarrow B$$

A agrees to buy gold from B in six months.

A has a problem:

“What happens if B disappears or goes bankrupt?”

B has the same concern about A.

An exchange uses a **clearing house**.

Conceptually:

$$A \leftrightarrow \boxed{\text{Clearing House}} \leftrightarrow B$$

A effectively has a contract with the clearing house.

B effectively has a contract with the clearing house.

This dramatically reduces direct counterparty-credit concerns.

The clearing house also requires traders to post **margin**.

Margin is money/collateral deposited to help ensure traders satisfy their obligations.

Chapter 2 goes much deeper into margin.

4. Electronic trading

Historically, futures exchanges often used **open outcry**.

You’ve probably seen old videos where traders stand in a pit shouting and using hand signals.

Modern exchanges are largely electronic.

A trader sends an order electronically and computers match compatible buy and sell orders.

This helped create:

Algorithmic trading: computers automatically execute trading strategies.

High-frequency trading: algorithms perform very large numbers of trades extremely quickly.

You don't need HFT mechanics for Chapter 1. Hull is mainly showing how derivatives markets evolved.

5. OTC — Over-the-Counter markets

Section 1.2 is very important.

Not every derivative trades on an exchange.

Suppose Bank A and Company B create a customized contract privately.

That's an **OTC derivative**.

OTC = negotiated outside a traditional exchange

Main participants include banks, large financial institutions, corporations and fund managers.

OTC contracts can usually be customized much more than exchange contracts.

For example:

“I need a currency hedge for exactly €13.47 million ending March 17.”

An OTC bank can potentially construct exactly that contract.

6. Exchange vs OTC

This distinction should be automatic for you.

Exchange-traded	OTC
Standardized	Often customized
Exchange trading	Negotiated privately/electronically between institutions
Clearing house typically involved	Bilateral clearing or CCP
Lower direct counterparty exposure because of clearing	Counterparty risk traditionally more important
Highly organized market	Flexible contracts
Usually many market participants	Often banks and institutional clients

A common interview question is:

Why would somebody use OTC rather than exchange-traded derivatives?

Answer:

Because OTC markets allow more customization.

Why prefer an exchange?

Because standardization, liquidity and clearing mechanisms can reduce trading and counterparty complications.

7. Counterparty risk

Suppose I owe you \$10 million under a derivative.

Everything looks good mathematically.

Then I go bankrupt.

Your contract might say you're owed \$10 million, but actually collecting it becomes another matter.

That's **counterparty credit risk**.

It means:

Risk that the other party fails to meet its obligations

This became particularly important during the financial crisis.

Hull discusses Lehman Brothers because Lehman had an enormous number of OTC contracts when it failed in 2008.

8. CCP — Central Counterparty

A **CCP** stands between counterparties.

Instead of:

$$A \leftrightarrow B$$

you effectively get:

$$A \leftrightarrow CCP \leftrightarrow B$$

The CCP becomes buyer to every seller and seller to every buyer.

This helps manage counterparty risk.

Following the financial crisis, regulators pushed many standardized OTC derivatives toward central clearing.

Hull highlights three regulatory directions after the crisis: more standardized OTC trading through organized platforms, greater use of CCPs and reporting transactions to central repositories.

These are historical/regulatory points in the chapter, not current-market statistics.

9. Systemic risk

This is different from ordinary counterparty risk.

Imagine:

$$Bank\ A \rightarrow Bank\ B \rightarrow Bank\ C \rightarrow Bank\ D$$

Bank A fails.

Bank B suffers enormous losses.

That causes B to fail.

Then C loses money.

Then C fails.

The problem spreads throughout the financial system.

That is **systemic risk**.

Systemic risk = risk that problems at one institution spread through the financial system

Think:

domino effect.

10. Market makers

Banks often act as **market makers** in OTC markets.

A market maker continuously provides prices at which it is willing to transact.

You'll encounter two prices:

$$\text{Bid} < \text{Offer}$$

The **bid** is the price at which the dealer is prepared to buy.

The **offer**, also called the **ask**, is the price at which the dealer is prepared to sell.

Example:

$$GBP/USD = 1.5541/1.5545$$

The bank buys GBP for:

$$1.5541$$

and sells GBP for:

$$1.5545$$

Therefore, if **you buy GBP from the bank**, you pay the offer:

$$1.5545$$

If **you sell GBP to the bank**, you receive the bid:

$$1.5541$$

The difference:

$$1.5545 - 1.5541$$

is the **bid-offer spread**.

11. Notional amount vs value

This is subtle and commonly misunderstood.

Suppose a derivative says:

“In one year, exchange \$100 million for another currency.”

Its **notional/principal amount** might therefore be:

\$100 million

That absolutely does **not** mean the derivative itself is worth \$100 million.

Perhaps changing market conditions mean the contract is currently only worth:

\$1 million

to one party.

So:

Notional amount $\neq$ market value of derivative

The giant OTC market numbers Hull quotes in Chapter 1 largely refer to notional amounts. They should not be interpreted as “this much money could be lost.”

12. Spot contract

Before forwards, understand **spot**.

A spot transaction essentially means:

Buy/sell now, with immediate or near-immediate settlement.

If gold is:

$$S_0 = \$2,000$$

and you buy gold today around that market price, you are dealing in the spot market.

The notation:

$$S_0$$

usually means today's spot price.

Later:

$$S_T$$

means the spot price at future time T .

This notation will appear everywhere in derivatives.

13. Forward contract

A **forward contract** is an agreement today to buy or sell something at a specified future date for a specified price.

Suppose:

Today we agree that I will buy one Bitcoin from you six months from now for \$70,000.

No matter what Bitcoin costs then, our contract says:

$$K = \$70,000$$

where K is the **delivery price**.

14. Long forward

The person agreeing to **buy** the asset in the future has the:

Long forward position

Think:

LONG = I will BUY.

Suppose:

$$K = 100$$

At maturity:

$$S_T = 130$$

You are allowed—indeed obligated—to buy something worth \$130 for \$100.

Gain:

$$130 - 100 = 30$$

So:

$$\boxed{\text{Long forward payoff} = S_T - K}$$

If:

$$S_T = 70$$

then:

$$70 - 100 = -30$$

You still have to buy for \$100.

That's why a forward is an **obligation**, not a choice.

15. Short forward

The person agreeing to **sell** has the:

$$\boxed{\text{Short forward position}}$$

Think:

SHORT = I will SELL.

The payoff is:

$$\boxed{K - S_T}$$

If:

$$K = 100, \quad S_T = 70$$

the short benefits:

$$100 - 70 = 30$$

because the short gets to sell something for \$100 when the market price is only \$70.

Notice:

$$(S_T - K) + (K - S_T) = 0$$

The gain of one party is the loss of the other.

This is why derivatives are often described as **zero-sum contracts**, before transaction costs and considering the broader economic benefits of hedging.

16. Payoff vs profit

This distinction becomes extremely important with options.

For a basic forward, entering the contract normally has essentially zero initial cost.

So in Chapter 1's simplified examples:

$$\text{profit} \approx \text{payoff}$$

For options this is not true because you must pay a premium.

Always distinguish:

Payoff

from

Profit = payoff - initial cost

17. Forward price

Suppose stock price:

$$S_0 = 60$$

Interest rate:

$$r = 5\%$$

No dividends.

Instead of buying the stock today for \$60, you could keep the money earning interest.

After one year:

$$60(1.05) = 63$$

So Hull's simple example says the fair one-year forward price should be:

$$\boxed{63}$$

Why?

Because otherwise arbitrage appears.

18. Forward price too high

Suppose market forward price is:

$$67$$

while fair value should be:

$$63$$

You can borrow \$60, buy the stock today and simultaneously agree to sell it forward for \$67.

After one year:

Loan repayment:

$$60(1.05) = 63$$

Receive from forward:

$$67$$

Risk-free difference:

$$67 - 63 = 4$$

That's arbitrage.

This simple idea becomes enormously important later in Hull.

19. Futures contract

A futures contract has the same fundamental idea as a forward:

Buy/sell an asset in the future at an agreed price.

But futures are normally:

exchange-traded and standardized

while forwards are normally:

OTC and customizable

Important comparison:

Forward	Futures
OTC	Exchange
Customized	Standardized
Counterparty risk traditionally more relevant	Clearing house
Usually settled according to contract terms	Marked to market through margin system
Private negotiation	Highly liquid market for many contracts

The detailed mechanics of futures—initial margin, maintenance margin, variation margin and daily settlement—belong mainly to Chapter 2.

20. Call option

Now we reach one of the most important concepts in the entire book.

A **call option** gives its holder:

the RIGHT to BUY

an underlying asset at a predetermined price.

Remember:

CALL $\rightarrow$ BUY.

Suppose you buy a call with:

$$K = 100$$

The strike price is \$100.

At expiration stock becomes:

$$S_T = 150$$

Of course you use your right to buy at \$100.

Value:

$$150 - 100 = 50$$

But if:

$$S_T = 80$$

why would you buy at \$100 when the market sells it for \$80?

You simply don't exercise.

Therefore:

$\text{Long call payoff} = \max(S_T - K, 0)$

21. Put option

A **put option** gives its holder:

the RIGHT to SELL

Remember:

PUT $\rightarrow$ SELL.

Suppose:

$$K = 100$$

and stock falls to:

$$S_T = 60$$

You can sell something worth \$60 for \$100.

Payoff:

$$100 - 60 = 40$$

If stock instead rises to \$130, there's no reason to sell it for \$100.

So:

$$\text{Long put payoff} = \max(K - S_T, 0)$$

22. Strike price

The **strike price**, also called **exercise price**, is the predetermined price at which the option allows you to buy or sell.

Usually written:

$$K$$

Example:

Apple \$200 call

means:

$$K = 200$$

The call holder has the right to buy Apple at \$200 according to the option's terms.

23. Expiration or maturity

An option does not last forever.

Its final date is the:

expiration date

or

maturity date

After expiration, the option disappears.

24. American vs European options

This distinction is extremely likely to be asked.

A **European option** can be exercised:

only at expiration

An **American option** can be exercised:

at any time up to and including expiration

Very important:

American and European do **not** describe geography.

An American-style option could trade outside America.

25. Options are rights, not obligations

This is probably the most important difference between options and forwards.

A forward says:

You **must** transact.

An option says:

You **may** transact.

Therefore:

Forward = obligation

whereas:

Option holder = right

But rights are valuable.

Therefore you must normally pay to acquire an option.

That payment is the:

option premium

26. Four fundamental option positions

Everything in basic options comes from four positions.

Position	Meaning
Long call	Buy the right to buy
Short call	Sell/write the right to buy
Long put	Buy the right to sell
Short put	Sell/write the right to sell

When you **buy** an option:

long option

When you **sell/write** an option:

short option

“Writing an option” means selling/creating the option position.

27. Long call profit

Suppose:

$$K = 100$$

Premium:

$$c = 7$$

Payoff:

$$\max(S_T - 100, 0)$$

Profit:

$$\boxed{\max(S_T - 100, 0) - 7}$$

If:

$$S_T = 130$$

profit:

$$30 - 7 = 23$$

If:

$$S_T = 80$$

payoff = 0.

Profit:

$$-7$$

Maximum loss:

$$7$$

Maximum gain:

theoretically unlimited

Break-even:

$$100 + 7 = 107$$

Notice something very important:

The call gets exercised whenever:

$$S_T > 100$$

but it only produces a net profit when:

$$S_T > 107$$

So:

Exercise condition $\neq$ profit condition

This directly answers the idea behind Hull Question 1.26.

You **can exercise an option and still lose money overall**.

28. Long put profit

Suppose:

$$K = 100$$

premium:

$$p = 6$$

Profit:

$$\boxed{\max(100 - S_T, 0) - 6}$$

If:

$$S_T = 70$$

profit:

$$30 - 6 = 24$$

Break-even:

$$100 - 6 = 94$$

Maximum loss:

$$6$$

For a normal stock that cannot fall below zero, maximum gross payoff occurs at $S_T = 0$:

$$100$$

so maximum profit:

$$100 - 6 = 94$$

29. Short call

If you sell/write the call, your profit is exactly the opposite of the long call:

$$\boxed{\text{Short call profit} = \text{premium} - \max(S_T - K, 0)}$$

Maximum gain:

$$\boxed{\text{premium}}$$

because the best thing that can happen is the option expires worthless.

Potential loss:

theoretically unlimited

because the stock could keep rising.

This is why selling naked calls can be very risky.

30. Short put

Short put profit:

$$\text{premium} - \max(K - S_T, 0)$$

Maximum gain:

$$\text{premium}$$

For a stock, the worst case is roughly:

$$S_T = 0$$

Maximum loss:

$$K - \text{premium}$$

per share.

If you sell a \$40 put for \$5:

$$\text{Maximum gain} = 5$$

and:

$$\text{Maximum loss} = 40 - 5 = 35$$

per share.

For a 100-share contract:

$$\text{\$500 max gain}$$

versus:

\$3,500 max loss

assuming the stock can fall to zero.

31. Contract multiplier

Hull's examples often use stock option contracts where one contract controls:

100 shares

So an option quoted at:

\$5

doesn't normally mean the contract costs \$5.

It means:

$$5 \times 100 = \$500$$

This is an extremely common calculation trap.

32. Bid and offer with options

Suppose option quote:

5.00/5.50

If you **buy**, you pay:

5.50

If you **sell**, you receive:

5.00

Remember:

Buyer pays offer.
Seller receives bid.

33. Strike price effects

Hull points out a basic relationship.

As strike K increases:

Call value generally:



because the right to buy becomes less attractive.

Put value generally:



because the right to sell at a higher price becomes more attractive.

Example:

A right to buy Apple at \$150 is more valuable than a right to buy it at \$250.

But a right to sell Apple at \$250 is more valuable than a right to sell it at \$150.

34. Time to maturity

Generally, Hull observes that options tend to become more valuable as the time available to exercise increases.

Intuition:

More time means more opportunity for useful price movements.

Chapter 11 treats these relationships much more precisely.

35. Hedgers

Now we reach Section 1.7.

A **hedger** already has some economic exposure and uses derivatives to reduce it.

Important:

Hedging is not primarily about making money

It is about reducing uncertainty.

36. Currency hedging with forwards

Suppose a US company knows:

“Three months from now I must pay £10 million.”

The company is worried GBP might rise.

If GBP rises, purchasing £10m becomes more expensive.

So the company should lock in the GBP purchase today.

It takes:

long GBP forward

because it will need to **buy GBP**.

The rule is:

Future foreign-currency payment → long forward

Now suppose the company will **receive** £10 million.

It worries GBP might fall.

So it locks in selling GBP:

short GBP forward

Rule:

Future foreign-currency receipt → short forward

Memorize the economic logic rather than the words.

37. Hedging does not guarantee a better outcome

Very important conceptual question.

Suppose you lock GBP at:

1.55

Then GBP falls to:

1.40

Without the hedge, you could have bought GBP much more cheaply.

So someone might say:

“The hedge was a mistake!”

Not necessarily.

The purpose was to remove uncertainty.

Insurance works similarly.

You don’t regret buying home insurance because your house didn’t burn down.

Therefore:

Successful hedge $\neq$ hedge that makes money
--

A successful hedge reduces the intended risk.

38. Hedging with options

Suppose you own:

1,000 shares

at \$28.

You’re afraid the price might collapse.

You could buy puts with:

$K = 27.50$

Now no matter how low the market falls, you have the right to sell around \$27.50.

That's called a:

protective put

Conceptually:

Stock + Put

behaves like stock with insurance.

Downside is limited while upside remains.

39. Forward hedge vs option hedge

This is one of the best questions in Chapter 1.

A **forward hedge** locks the price.

A **put/call option hedge** creates insurance.

Suppose you need GBP.

Forward:

You lock an exchange rate

Whether GBP rises or falls, you use the agreed rate.

Option:

You protect yourself if GBP rises but can benefit if GBP falls.

That flexibility costs money.

So:

Forward hedge	Option hedge
Locks price	Creates protection
Removes upside and downside	Protects downside while keeping favorable moves
Usually no initial premium	Premium required
Obligation	Right

Think:

Forward = lock

Option = insurance

40. Which currency option should you buy?

This appears in Hull's questions.

If you will **receive foreign currency**, you're worried its value will fall.

You will need to **sell** the foreign currency.

So buy a:

put on the foreign currency

because a put gives you the right to sell.

If you will **pay foreign currency**, you're worried it will rise.

You need to buy it.

So buy a:

call on the foreign currency

because a call gives you the right to buy.

This rule is worth memorizing.

41. Speculators

A speculator deliberately takes market risk because they have a view.

Hedger:

"GBP might rise and hurt my business."

Speculator:

"I think GBP will rise, and I want to profit from it."

The derivative contract can be identical.

The purpose is what changes.

42. Speculation using futures

Suppose you believe GBP will rise.

One possibility:

Buy:

£250,000

directly.

That requires a huge amount of cash.

Another possibility:

Buy GBP futures representing:

£250,000

You may only need to deposit a comparatively small amount as margin.

You therefore control a large economic position using much less initial cash.

That's:

leverage

43. What is leverage?

Leverage means:

small amount of capital controls a much larger economic exposure

Suppose you put up:

\$20,000

but your futures exposure represents:

\$400,000

A small percentage change in the underlying can create a very large percentage gain or loss relative to your \$20,000.

Leverage magnifies:

profits AND losses

This second part is crucial.

44. Speculating with options

Suppose stock:

$$S_0 = 20$$

You have:

\$2,000

Strategy A:

Buy 100 shares.

Strategy B:

Buy 2,000 call options costing \$1 each.

Suppose stock rises to:

27

Shares:

$$100(27 - 20) = 700$$

Call has:

$$K = 22.50$$

Payoff per option:

29

$$27 - 22.50 = 4.50$$

Total payoff:

$$2000(4.50) = 9000$$

Initial cost:

$$2000$$

Profit:

$$\$7,000$$

The same \$2,000 produced much more exposure.

That's option leverage.

But if stock collapses, the calls could expire worthless and the entire \$2,000 premium could be lost.

45. Futures leverage vs option leverage

Both provide leverage.

But there is a major difference.

With a **long option**:

$\text{maximum loss} = \text{premium}$
--

With futures:

Loss can become very large because the contract creates an obligation.

So for a bullish speculator:

Long futures:

large upside, large downside

Long call:

large upside, loss limited to premium

But remember: the option protection isn't free. You paid the premium.

46. Arbitrage

This is Section 1.9 and is absolutely fundamental to the rest of Hull.

An arbitrage opportunity allows you to construct transactions that lock in profit without taking the corresponding market risk.

Simplified definition:

Arbitrage = risk-free profit from inconsistent prices

47. Hull's dual-market arbitrage example

Imagine the same stock trades:

New York:

\$150

London:

£100

Exchange rate:

\$1.53/£

The London stock is effectively worth:

$$100(1.53) = \$153$$

Yet New York sells the identical asset for:

\$150

So:

Buy in New York:

−\$150

Sell equivalent stock in London:

+\$153

Profit:

\$3

per share before transaction costs.

For 100 shares:

\$300

You didn't need the stock price to rise or fall.

You exploited an inconsistency.

48. Why arbitrage disappears

Once arbitrageurs notice:

$NY = \$150$

while:

$London = \$153$

they buy in New York.

Buying pressure pushes New York price upward.

They sell in London.

Selling pressure pushes London price downward.

Eventually prices converge.

Therefore arbitrage trading itself helps enforce:

consistent market prices

This leads to the fundamental **no-arbitrage principle** used throughout the book.

49. No-arbitrage principle

Much of modern derivatives pricing is built around this idea:

Two portfolios producing exactly the same future cash flows should have the same price today.

Otherwise:

Buy the cheaper one.

Sell the expensive one.

Lock in the difference.

This is the seed from which you'll later get forward pricing, put-call parity, binomial option pricing and Black-Scholes reasoning.

So when Hull says:

No arbitrage

take it very seriously.

It becomes one of the central assumptions of the book.

50. Hedger vs speculator vs arbitrageur

A good way to distinguish them:

Situation	Trader
Airline buys oil futures because it needs fuel	Hedger
Trader buys oil futures because they expect oil to rise	Speculator
Trader simultaneously buys cheap oil exposure and sells identical overpriced exposure	Arbitrageur

Notice something important:

The financial instrument doesn't tell you the trader type.

Purpose does.

The same futures contract may be used for hedging or speculation.

That directly answers one of Hull's practice questions.

51. Why derivatives are dangerous

Section 1.10 deals with an important non-mathematical issue.

Derivatives are extremely flexible.

That means an employee who is supposed to be:

hedging

can secretly start:

speculating

Because derivatives can involve enormous leverage, losses can grow very quickly.

Hull discusses Jérôme Kerviel at Société Générale as an example of a trader whose unauthorized positions led to enormous losses.

The broader lesson isn't simply "derivatives are dangerous."

It's:

leverage + weak risk controls = potentially enormous losses

Financial institutions therefore use position limits, risk limits, collateral rules, independent risk management, daily monitoring and other controls.

Hull's deeper risk-management lesson is essentially:

What can go wrong, and how much could we lose if it does?

52. Selling a call vs buying a put

Hull explicitly asks this.

They are **not the same thing**.

Buying a put means:

I have the right to sell at K .

You pay premium.

Your downside is limited to the premium.

Selling a call means:

Someone else has the right to buy from me at K .

You receive premium.

But if the asset rises dramatically, your losses can be enormous.

Their payoff shapes and risk profiles are very different.

53. Forward vs call with the same K

Another Hull question:

Suppose:

$$K = 50$$

Long forward:

You **must** buy for \$50.

Long call:

You **may** buy for \$50.

If:

$$S_T = 80$$

both positions benefit from being able to transact around \$50.

But if:

$$S_T = 20$$

forward:

You still buy for \$50.

Loss:

$$20 - 50 = -30$$

Call:

You simply don't exercise.

Your loss is limited to the premium.

That's why the call costs money upfront while entering a forward usually does not.

54. Exercise vs profitability

This distinction deserves repeating because Hull tests it.

Suppose:

$$K = 30$$

Call premium:

$$3$$

At maturity:

$$S_T = 31$$

Should you exercise?

Yes.

You buy at \$30 something worth \$31.

Exercise payoff:

$$1$$

But total profit:

$$1 - 3 = -2$$

So you exercise while still losing \$2 overall.

Why?

Because the \$3 premium has already been paid.

At expiration, that payment is a sunk cost.

Your decision is simply:

$$31 > 30?$$

Yes $\rightarrow$ exercise.

Break-even is:

$$K + \text{premium} = 33$$

Therefore:

$$30 < S_T < 33$$

means:

exercise the call but still lose money overall.

Excellent interview question.

55. Maximum gain and loss summary

This table is worth knowing cold.

Position	Maximum gain	Maximum loss
Long call	Unlimited	Premium
Short call	Premium	Unlimited
Long put on stock	Approximately $K - \text{premium}$	Premium
Short put on stock	Premium	Approximately $K - \text{premium}$
Long forward	Very large/unbounded as asset rises	Large as asset falls
Short forward	Large as asset falls	Unbounded as asset rises

For puts, those limits assume a normal stock cannot fall below zero.

56. Why stock options don't normally fund the company

Hull asks something like:

A newly issued stock provides funds to the company. Does a stock option do the same?

Usually, when exchange-traded stock options are bought and sold, the transaction occurs between market participants.

The underlying company isn't receiving the option premium.

Compare:

Company issues new shares:

$$Investor \rightarrow Company$$

Company receives capital.

Investor buys an exchange-traded option:

$$Investor \rightarrow Other\ market\ participant$$

The company normally isn't receiving that money.

Employee stock options or warrants issued directly by a corporation are a different situation.

57. Why options can be considered insurance

Suppose:

$$S_0 = 100$$

You own the stock.

You buy a put:

$$K = 90$$

If stock rises to \$150, you keep the upside.

If stock falls to \$20, the put lets you sell around \$90 according to its terms.

Your downside is protected.

The put premium is analogous to an:

insurance premium

So:

Stock + put = insured stock position

58. Forward payoff questions — fast method

Whenever Hull gives you a forward question, identify three things:

$$K, \quad S_T, \quad Q$$

where Q is quantity.

Long forward:

$Q(S_T - K)$

Short forward:

$Q(K - S_T)$

Example:

Short £100,000.

$$K = 1.5000$$

At maturity:

$$S_T = 1.4900$$

Profit:

$$100,000(1.5000 - 1.4900)$$

$$= \$1,000$$

If:

$$S_T = 1.5200$$

profit:

$$100,000(1.5000 - 1.5200)$$

$$= -\$2,000$$

That solves the style of Question 1.5 immediately.

59. Futures payoff questions

At the Chapter 1 level, use basically the same price-change logic.

Suppose you short cotton futures at:

50 cents/lb

Quantity:

50,000 lb

Final price:

48.2 cents

Because you're short:

$$(0.500 - 0.482)(50,000)$$

$$= 0.018(50,000)$$

$$= \boxed{\$900 \text{ gain}}$$

If price becomes:

51.3 cents

$$(0.500 - 0.513)(50,000)$$

$$= -\$650$$

Short positions benefit from falling prices.

60. The mental model that makes Chapter 1 easy

Before solving any problem, ask these questions in order:

1. **What is the underlying?**
2. **What does the person need or believe will happen?**
3. **Are they hedging, speculating, or arbitraging?**
4. **Is the contract a forward/future or an option?**
5. **Long or short?**
6. **If option: call or put?**
7. **What are S_T , K , quantity and premium?**
8. **Am I calculating payoff or profit?**
9. **Is there an obligation or merely a right?**
10. **If two equivalent things have different prices, is there an arbitrage?**

If you can answer those ten questions, almost every Chapter 1 problem becomes mechanical.

The formulas you should know without thinking

Forward

Long:

$$\boxed{S_T - K}$$

Short:

$$\boxed{K - S_T}$$

For Q units:

$$Q(S_T - K)$$

or

$$Q(K - S_T)$$

Call

Long-call payoff:

$$\boxed{\max(S_T - K, 0)}$$

Long-call profit:

$$\boxed{\max(S_T - K, 0) - c}$$

Short-call profit:

$$\boxed{c - \max(S_T - K, 0)}$$

Put

Long-put payoff:

$$\boxed{\max(K - S_T, 0)}$$

Long-put profit:

$$\boxed{\max(K - S_T, 0) - p}$$

Short-put profit:

$$\boxed{p - \max(K - S_T, 0)}$$

where:

S_T = underlying price at maturity

K = strike/delivery price

c = call premium

p = put premium

The most important verbal answers to memorize conceptually

What is a derivative?

A financial instrument whose value depends on another underlying variable.

Forward vs future?

Both create an obligation to transact in the future. Forwards are typically customized OTC contracts; futures are standardized exchange-traded contracts supported by a clearing and margin system.

Forward vs option?

A forward creates an obligation. An option gives the holder a right but not an obligation and therefore requires a premium.

Call vs put?

Call = right to buy. Put = right to sell.

Long vs short?

Long forward = agree to buy. Short forward = agree to sell. Long option = buy the option. Short option = sell/write it.

Hedger?

Uses derivatives to reduce an existing risk.

Speculator?

Uses derivatives to deliberately take a market view and profit if that view is correct.

Arbitrageur?

Uses inconsistent prices in different markets to construct offsetting transactions and lock in a profit.

What is leverage?

Controlling a large economic exposure using a relatively small initial investment.

Why hedge if hedging can produce a worse final outcome?

Because hedging's purpose is to reduce uncertainty, not guarantee the best ex-post result.

Why use an option instead of a forward for hedging?

A forward fixes the price completely. An option protects against adverse movements while preserving the benefit of favorable movements, in exchange for a premium.

What is systemic risk?

The risk that failure of one financial institution creates losses that spread through the financial system.

Why are arbitrageurs useful?

Their trading pushes inconsistent prices back toward relationships consistent with no-arbitrage.

What I would concentrate on for your internship questions

Based on the actual end-of-chapter questions in your copy, make absolutely sure you can solve the following without hesitation:

1. long forward vs short forward;
2. hedging vs speculation vs arbitrage;
3. forward vs call;
4. buying put vs selling call;
5. forward/futures payoff calculations;
6. the four option positions;
7. exercise vs profit;
8. maximum option gain/loss;
9. exchange vs OTC;
10. bid vs offer;
11. using forwards/options to hedge currency payments and receipts;
12. futures/options leverage;
13. protective puts as insurance;
14. zero-sum nature of derivatives;
15. detecting simple arbitrage;
16. explaining why arbitrage removes price discrepancies;
17. constructing payoff expressions with $\max()$.

Those areas cover essentially the conceptual machinery Hull expects you to have after Chapter 1. Your uploaded copy's Chapter 1 also ends with practice and further questions specifically targeting these points (printed pp. 41–44).

If you want, I can next take **all Chapter 1 practice questions 1.1–1.39 one by one**, solve them, and explain *why* each answer works rather than just giving the solution.